

BEAUSOLEIL, ON, Canada

WMO: 714325

Lat: 44.850N

Lon: 79.870W

Elev: 183

StdP: 99.15

Time Zone: -5.00 (NAE)

Period: 94-07

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.5	-20.9	-28.1	0.3	-23.6	-24.5	0.4	-20.6	7.3	-6.2	6.7	-6.4	1.6	340	0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	29.9	23.3	28.1	22.1	26.5	21.3	24.4	28.1	23.3	26.6	22.2	25.1	2.3	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	18.5	26.4	22.2	17.3	25.1	21.2	16.3	24.0	74.6	28.2	70.3	27.0	66.3	25.1	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.2	5.4	4.8	DB	-28.1	32.8	4.7	1.8	-31.5	34.1	-34.2	35.1	-36.9	36.1	-40.3	37.4	(4)
			WB	-28.9	26.4	4.6	1.2	-32.2	27.3	-34.9	28.0	-37.5	28.7	-40.8	29.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.0	-7.9	-7.3	-2.3	5.1	11.8	17.7	20.4	19.8	16.1	9.1	3.2	-3.0
	DBStd	11.15	7.49	5.82	5.68	4.76	4.50	4.07	3.12	3.05	3.98	4.64	4.83	5.68
	HDD10.0	2306	557	484	384	160	33	1	0	0	3	74	208	402
	HDD18.3	4364	815	717	641	398	208	59	15	19	87	289	454	660
	CDD10.0	1197	0	0	1	12	88	232	323	304	187	46	3	1
	CDD18.3	215	0	0	0	1	6	41	81	65	21	2	0	0
	CDH23.3	1366	0	0	0	9	53	304	499	367	122	11	0	0
	CDH26.7	331	0	0	0	1	11	83	116	97	21	2	0	0
Wind	WSAvg	2.1	2.4	2.3	2.2	2.4	2.1	1.8	1.7	1.7	1.9	2.1	2.5	2.5
Precipitation	PrecAvg	921	93	59	55	66	82	80	83	62	83	84	90	93
	PrecMax	1035	151	112	97	135	166	161	155	112	128	150	165	142
	PrecMin	742	52	21	15	20	32	22	29	17	26	20	33	39
	PrecStd	81	27	23	23	30	29	31	30	23	28	37	35	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	8.3	16.3	24.2	27.9	31.4	31.7	32.5	29.5	24.7	16.3	11.5
		MCWB	7.0	5.7	13.2	16.4	20.2	24.2	25.2	23.7	22.0	18.5	13.6	10.0
	2%	DB	5.4	4.9	10.7	18.7	24.4	29.1	29.4	29.2	26.6	20.9	13.7	8.4
		MCWB	4.4	3.2	8.7	12.3	18.5	22.8	23.5	22.9	20.9	17.5	11.7	7.0
	5%	DB	3.4	2.8	7.4	15.3	21.6	27.0	27.8	27.0	24.5	18.1	11.5	5.9
		MCWB	2.7	1.7	4.9	10.4	16.4	21.9	22.1	21.7	19.5	15.4	9.7	4.7
	10%	DB	1.5	1.2	4.9	12.2	19.1	24.6	26.2	25.0	22.5	15.9	9.7	3.9
		MCWB	0.9	0.3	3.3	8.0	15.0	20.3	21.3	20.5	18.8	13.7	8.3	2.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.5	6.0	13.9	17.2	21.8	25.2	26.3	25.6	23.2	20.2	14.6	10.6
		MCDB	7.8	7.0	15.5	21.5	25.2	29.1	30.0	28.7	27.5	23.0	15.9	11.8
	2%	WB	4.5	3.6	8.8	13.8	19.5	23.7	24.6	23.7	21.7	17.7	12.2	7.2
		MCDB	5.1	4.4	10.9	18.2	23.3	27.8	28.0	27.5	24.8	19.8	13.4	8.2
	5%	WB	2.7	1.8	5.3	10.8	17.6	22.5	23.4	22.6	20.5	15.7	10.0	4.8
		MCDB	3.4	2.7	6.7	14.4	20.1	26.0	26.3	25.6	23.6	17.9	11.1	5.5
	10%	WB	1.0	0.4	3.4	8.6	15.7	21.1	22.3	21.6	19.2	13.9	8.3	3.0
		MCDB	1.5	1.2	4.7	11.5	18.6	23.7	24.9	24.0	21.8	15.5	9.4	3.8
Mean Daily Temperature Range	5% DB	MDBR	8.8	10.3	9.6	9.9	10.2	10.1	9.6	9.3	9.3	8.0	6.8	7.1
		MCDBR	7.8	10.4	11.3	15.2	13.9	12.9	11.3	11.6	12.0	11.5	9.7	8.3
		MCWBR	7.1	8.9	8.6	9.7	8.2	7.2	6.0	5.6	6.5	7.6	8.1	6.9
	5% WB	MCDBR	7.7	9.3	10.5	14.0	12.3	11.8	10.0	9.6	10.5	10.9	9.1	7.6
		MCWBR	7.2	8.2	8.7	9.8	8.1	7.3	6.1	5.5	6.6	7.7	8.2	6.9
Clear-Sky Solar Irradiance	taub	0.287	0.296	0.317	0.352	0.384	0.405	0.408	0.401	0.365	0.345	0.313	0.286	
	taud	2.396	2.338	2.387	2.369	2.350	2.311	2.329	2.340	2.431	2.464	2.499	2.458	
	Ebn at Noon	832	891	919	909	886	865	857	850	856	825	799	799	
	Edh at Noon	74	94	103	115	121	126	123	117	99	83	68	63	
All-Sky Solar Radiation	RadAvg	1.30	2.20	3.56	4.52	5.38	5.87	5.94	5.09	3.98	2.32	1.34	0.94	
	RadStd	0.15	0.27	0.37	0.42	0.53	0.44	0.37	0.28	0.38	0.25	0.19	0.11	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page